



BMBank

Short-term bank node and UI management project for the
subject of PV

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1. Introduction

This documentation describes the development and features of the BMBank project – a lightweight bank management system consisting of a TCP-based server (bank node) and a WPF UI client. The system allows users to manage bank accounts, track client activities, and interact with the bank node via TCP commands. Developed using .NET 8.0 and WPF, it emphasizes data integrity, persistence via Microsoft SQL Server, and modular, maintainable code.

BMBank – Admin UI

Clients Accounts

IP Address	Last Command	Timestamp	Status
127.0.0.1:51188	AC	1/27/2026 7:24:00 PM	OK
127.0.0.1:51188	BN	1/27/2026 7:23:50 PM	OK
127.0.0.1:51188	BA	1/27/2026 7:23:43 PM	OK
127.0.0.1:51188	BA	1/27/2026 7:23:42 PM	OK
127.0.0.1:63499	BA	1/27/2026 7:21:40 PM	OK
127.0.0.1:63499	BA	1/27/2026 7:21:34 PM	OK
127.0.0.1:63499	BA	1/27/2026 7:21:33 PM	OK
127.0.0.1:63499	BA	1/27/2026 7:21:32 PM	OK
127.0.0.1:63499	BA	1/27/2026 7:21:31 PM	OK
127.0.0.1:63499	BA	1/27/2026 7:21:30 PM	OK
127.0.0.1:63499	BA	1/27/2026 7:21:30 PM	OK
127.0.0.1:63499	BA	1/27/2026 7:21:29 PM	OK
127.0.0.1:63499	BA	1/27/2026 7:21:28 PM	OK
127.0.0.1:63499	BA	1/27/2026 7:21:10 PM	OK
127.0.0.1:63499	BA	1/27/2026 7:21:09 PM	OK
127.0.0.1:63499	BA	1/27/2026 7:21:03 PM	OK
127.0.0.1:63499	RP	1/27/2026 7:20:33 PM	ERROR
127.0.0.1:63499	BN	1/27/2026 7:20:28 PM	OK
127.0.0.1:63699	BN	1/27/2026 7:16:01 PM	OK
127.0.0.1:63696	BN	1/27/2026 7:15:58 PM	OK

OFFLINE

Picture 1: UI

2. User specifications and requirements

The system is designed to provide a centralized interface for bank administrators to perform the following tasks:

- Manage bank accounts: create, deposit, withdraw, check balance, remove accounts.
- Monitor client activity: view last executed commands, client IPs, execution timestamp, and command result.
- Maintain data persistence: store account and client data in SQL Server.
- Node-to-node communication: P2P network commands for bank discovery and distributed operations.
- Periodic data refresh in UI to reflect real-time server state.

3. Application architecture

Server (Bank Node)

- **Server class:** TCP server handling incoming client connections.
- **ClientSession:** Represents a session with a connected client, processing commands asynchronously.
- **CommandHandler:** Maps incoming command keys to ICommand implementations and executes them.
- **BankNetworkExplorer & BankProxyClient:** Handle discovery of other bank nodes and forwarding commands across the P2P network.
- **DAO layer (AccountDAO, ClientCommandLogDAO):** Access SQL Server database, manage accounts, transactions, and logging.
- **Logger & SafeExecutor:** Provide thread-safe logging and standardized error handling.
- **NetworkConfig & IPAddressObtainer:** Configure ports, network prefixes, and obtain local IP addresses.

UI (WPF Client)

- **MainViewModel:** Implements MVVM, exposes active accounts, closed accounts, and recent client activity.
- **BankTcpServices:** Provides TCP communication with the server node, checks server status, retrieves runtime info.
- **DatabaseService:** Read-only access to SQL Server, retrieves data for UI.
- **AccountInfo & ClientInfo:** Data models representing accounts and client activity for UI binding.

4. Behavioral logic

Server

1. Listens on TCP port for incoming connections.
2. Validates and executes commands (locally or proxied).
3. Logs command executions with timestamp, client IP, and status.
4. Supports P2P interactions for bank discovery and “robbery plan” commands.

UI

1. Reads App.config for server IP, port, and SQL credentials.
2. Connects to server via TCP.
3. Refreshes UI periodically with:
 - Server status (ONLINE/OFFLINE)
 - Active accounts
 - Closed accounts
 - Client activity
4. Displays data via ObservableCollection and WPF binding.

5. Database system description

SQL Server backend:

- **Tables:** Accounts, ClientCommandLog, plus views view_active_accounts, view_closed_accounts, view_ui_client_activity.
- **Account entity:** account_number, balance, status, created_at, optional closed_at.
- **ClientCommandLog:** client_ip, command, arguments, result status, timestamp.
- Operations wrapped in transactions for consistency.

6. Server and UI configuration

Connection Parameters

Settings are managed in App.config under the <appSettings> section:

Server:

- DataSource: The IP or server name of the SQL instance.
- Database: The target database name.
- Login/Password: SQL Server authentication credentials.
- BankPortFrom: Starting TCP port number used when scanning the network for other bank nodes
- BankPortTo: Ending TCP port number used when scanning the network for other bank nodes
- BankNetworkPrefix: IPv4 network prefix used for discovering other bank nodes in the P2P network

UI:

- DataSource: The IP or server name of the SQL instance.
- Database: The target database name.
- Login/Password: SQL Server authentication credentials.
- ServerIp: IPv4 address of the bank server the UI connects to via TCP
- ServerPort: TCP port on which the bank server is listening

Supplementary Setup Instructions

Beyond the README, ensure that:

1. File Permissions: The application has written access to its own directory to create the Logs/ folder.
2. SQL Permissions: The database user must have SELECT, INSERT, UPDATE, and DELETE permissions, as well as permission to execute VIEW queries.

More information about the project configuration can be found in the provided README file.

7. Error handling

- **SafeExecutor**: Wraps critical code in try-catch-finally for safe execution.
- **Logger**: Thread-safe logging, daily log rotation, severity levels (Info, Warning, Error).
- Provides detailed records of server operations, client commands, and system errors.

8. Commands and network operations

Name	Code	Description
Bank code	BC	Returns local bank code (IP address)
Account create	AC	Create new account, returns account number + local IP
Account deposit	AD	Deposit to account
Account withdrawal	AW	Withdraw from account
Account balance	AB	Get account balance
Account remove	AR	Remove account (local only)
Bank total amount	BA	Get total bank amount
Bank number of clients	BN	Get number of clients
Robbery plan	RP	Robbery plan (simulated assignment command)

9. Third party libraries

The project maintains a minimal footprint by utilizing only essential libraries:

- Microsoft.Data.SqlClient: Enables the use of advanced features such as SqlTransaction, SqlDataReader, and asynchronous database commands.
- System.Configuration.ConfigurationManager: Used within the SqlConnectionFactory to securely retrieve sensitive connection strings and provider settings from the App.config file.
- WPF Toolkit / Standard Libraries: The application relies on the standard .NET 8.0 libraries for UI rendering (WPF), data binding via INotifyPropertyChanged, and collection management through ObservableCollection.

10. Project summary and additional information

BMBank demonstrates:

- TCP client-server communication
- MVVM-based WPF UI
- SQL Server persistence with DAO pattern
- P2P bank node discovery
- Thread-safe logging and robust error handling

Developed as a school project for PV subject. Serves as foundation for further enhancements.

Střední průmyslová škola elektrotechnická website: <https://www.spsejecna.cz/>

Github repository: <https://github.com/LupusLudit/BMBank>

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